QUASI-COMPACT OPENS OF AN INVERSE LIMIT OF SPECTRAL SPACES DESCEND TO A FINITE STAGE

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Allowed standard prerequisites (editorial).

- The statement that constructible subsets of the limit come from a finite stage. Invoked inside the counted proof (source lines 5211-5213). Descends the subset to a stage.
- The statement on differences of pullbacks in such a limit. Invoked inside the counted proof (source lines 5214-5216). Provides the nonemptiness criterion for the differences appearing in the statement.

Source reference. Original lemma lemma-inverse-limit-spectral-spaces-constructible: sources/stacks-topology/pinned-original-source.tex, source lines 5079–5079. Navigation pointer only: the counted text refers to the Stacks item lemma-inverse-limit-spectral-spaces-constructible, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-inverse-limit-spectral-spaces-equal: sources/stacks-topology/pinned-original-source.tex, source lines 5053–5053. Navigation pointer only: the counted text refers to the Stacks item lemma-inverse-limit-spectral-spaces-equal, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $\mathcal{I}$ be a cofiltered index category. Let $i \mapsto X_i$ be a diagram of spectral spaces such that for $a : j \to i$ in $\mathcal{I}$ the corresponding map $f_a : X_j \to X_i$ is spectral. Set $X = \lim X_i$ and denote $p_i : X \to X_i$ the projection.*

(1) *Given any quasi-compact open $U \subset X$ there exists an $i \in \mathrm{Ob}(\mathcal{I})$ and a quasi-compact open $U_i \subset X_i$ such that $p_i^{-1}(U_i) = U$.*

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(2) *Given* $U_i \subset X_i$ *and* $U_j \subset X_j$ *quasi-compact opens such that* $p_i^{-1}(U_i) \subset p_j^{-1}(U_j)$ *there exist* $k \in \mathrm{Ob}(\mathcal{I})$ *and morphisms* $a : k \to i$ *and* $b : k \to j$ *such that* $f_a^{-1}(U_i) \subset f_b^{-1}(U_j)$.
(3) *If* $U_i, U_{1,i}, \ldots, U_{n,i} \subset X_i$ *are quasi-compact opens and* $p_i^{-1}(U_i) = p_i^{-1}(U_{1,i}) \cup \ldots \cup p_i^{-1}(U_{n,i})$ *then* $f_a^{-1}(U_i) = f_a^{-1}(U_{1,i}) \cup \ldots \cup f_a^{-1}(U_{n,i})$ *for some morphism* $a : j \to i$ *in* $\mathcal{I}$.
(4) *Same statement as in (3) but for intersections.*

Proof. Part (1) is a special case of Lemma lemma-inverse-limit-spectral-spaces-constructible. Part (2) is a special case of Lemma lemma-inverse-limit-spectral-spaces-equal as quasi-compact opens are both open and closed in the constructible topology. Parts (3) and (4) follow formally from (1) and (2) and the fact that taking inverse images of subsets commutes with taking unions and intersections. □